

01 Introduction to HTML

Markup basics · Document structure · Elements, attributes and nesting

What HTML is

HTML stands for **HyperText Markup Language**. It is a **markup language**, not a programming language. Its job is not to run logic but to give the content **meaning and structure** — this is a heading, this is a paragraph, this is a list, this is an image. The browser reads this structure and renders it on the screen.

- **HyperText** — documents linked to one another through hyperlinks.
- **Markup** — marking up content with tags so that its meaning is defined.
- HTML has no loops, conditions or variables, which is exactly why it is not called a programming language.

Basic document structure

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Page Title</title>
  </head>
  <body>
    <h1>Hello World</h1>
  </body>
</html>
```

Part	Purpose
<code><!DOCTYPE html></code>	Tells the browser that the document is HTML5. It is a declaration, not a tag. Remove it and the browser falls back to <i>quirks mode</i> , where the box model and some CSS rules behave in a legacy manner.
<code><html lang="en"></code>	The root element. The <code>lang</code> attribute tells screen readers and search engines which language the page is written in.
<code><head></code>	Metadata — title, character set, viewport, stylesheet links and SEO tags. This content is not displayed on the page.
<code><meta charset="UTF-8"></code>	Character encoding. Without it, regional language text, emoji and special characters may render as garbled symbols.
<code><body></code>	The actual visible content — headings, paragraphs, images and everything else the user sees.

Elements, tags and attributes

```
<a href="https://example.com" target="_blank">Visit</a>
|_| |_____| |_| |_|
tag      attributes      content  closing tag
```

- An **element** is the complete unit: opening tag, content and closing tag.
- An **attribute** is always written inside the opening tag, in the form `name="value"`.
- **Void (empty) elements** have no closing tag — `<img>`, `<br>`, `<hr>`, `<input>`, `<meta>` and `<link>`.
- **Nesting** must be closed in the correct order. `<p><strong>text</strong></p>` is valid; `<p><strong>text</p></strong>` is not.
- Comments are written as `<!-- this is a comment -->`. The browser ignores them, but they remain **visible in view-source**, so credentials or secrets must never be placed there.

Behaviour that surprises most beginners

- **Whitespace collapses.** Ten spaces or five line breaks in the source will display as a single space. Spacing is the job of CSS, not of extra whitespace.
- **Tag names are case-insensitive.** `<DIV>` and `<div>` both work, but lowercase is the accepted industry convention.
- **HTML is error tolerant.** A syntax mistake does not crash the page; the browser silently repairs the markup in its own way, which usually results in an unexpected layout.
- **Block versus inline.** Elements such as `div`, `p` and `h1` start on a new line and take the full available width, whereas `span`, `a` and `strong` occupy only as much width as their content.

COMMON MISTAKES

Placing `<title>` inside `<body>` · forgetting closing tags · omitting quotes around attribute values · saving the file as `index.txt` instead of `index.html`
· assuming CSS and JavaScript are part of HTML itself.

QUICK RECAP

- HTML gives structure and meaning; CSS handles presentation; JavaScript handles behaviour.
- Every page needs a DOCTYPE, a `lang` attribute, a character set and a title.
- Void elements take no closing tag, and nesting must always be closed inside out.